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Land Use Intensity-Specific Global Characterization Factors to Assess Product Biodiversity Footprints

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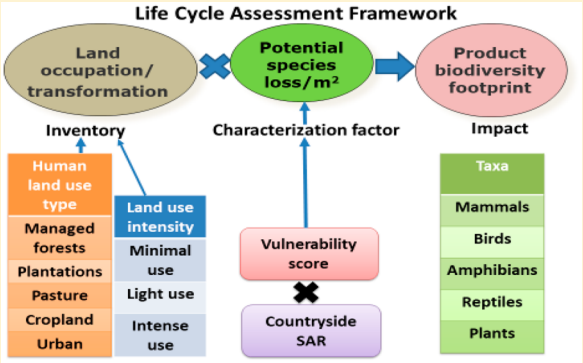
Supporting Information

ABSTRACT: The UNEP-SETAC life cycle initiative recently recommended use of the countryside species−area relationship (SAR) model to calculate the characterization factors (CFs; potential species loss per m^2) for projecting the biodiversity impact of land use associated with a products' life cycle. However, CFs based on this approach are to date available for only six broad land use types without differentiating between their management intensities and have large uncertainties that limit their practical applicability. Here we derive updated CFs for projecting potential species losses of five taxa resulting from five broad land use types (managed forests, plantations, pasture, cropland, urban) under three intensity levels (minimal, light, and intense use) in each of the 804 terrestrial ecoregions. We utilize recent global land use intensity maps and International Union for Conservation of Nature (IUCN) habitat classification scheme to parametrize the SAR model. As a case study, we compare the biodiversity impacts of 1 m^3 of wood produced under four different forest management regimes in India and demonstrate that the new land use intensity-specific CFs have smaller uncertainty intervals and are able to discern the impacts of intensively managed land uses from the low intensity regimes, which has not been possible through previous CFs.



INTRODUCTION

Environmentally conscious purchasing behavior by individual consumers, businesses, and nations can contribute substantially toward mitigating the ongoing biodiversity crisis.[1] Informing consumers about the biodiversity impacts "hidden" in the life cycle of products they consume is an important step toward transforming current consumption patterns.[2−8] Life cycle assessment (LCA) is increasingly used for estimating and comparing the "cradle to grave" environmental impacts of products and commodities.[9] In LCA, biodiversity impacts of land use are calculated by multiplying inventory (i.e., land occupation in m^2·year and land transformation in m^2) with characterization factors (CFs, i.e., the factors indicating biodiversity damage caused by unit area of a particular land use).[4]

A suite of CFs have been proposed to assess the species loss within a product's life cycle, using different ecological models and varying taxonomic coverage and spatial resolution.[5,9,10] Different results are obtained when comparing the products depending upon which CFs are employed. A lack of consensus on which CFs best reflect the damage inflicted by a given product system has, thus, hampered the application of these methods to inform decision makers and stakeholders.[11]

To address this challenge, the land use task force of the United Nations Environment Programme/Society of Environmental Toxicology and Chemistry (UNEP-SETAC) Life Cycle Initiative was established to provide guidance and build scientific consensus on CFs for assessing land use impacts on biodiversity.[12] Over 2015−2017, the initiative organized discussions between experts and stakeholders[13] as well as conducted a critical review of the existing framework and biodiversity models.[5] It concluded that an acceptable model should incorporate both the intensity of human land use as well as the vulnerability of the species group in the region where the land use is taking place (e.g., if the region in question holds many endemic or threatened species). The initiative culminated in a Pellston Workshop, which provisionally recommended the CFs calculated by Chaudhary et al.[4] to assess land use driven biodiversity impacts within LCA (see UNEP report).[14]

The CFs calculated by Chaudhary et al.[4] represent biodiversity damage resulting from global land use (in potential species lost per m^2) for four taxa: mammals, birds, amphibians, and reptiles. The CFs assess the potential damage caused by a unit area of six different land use types (annual crops, permanent crops, pasture, intensive forestry, extensive forestry, and urban land) in each of the 804 terrestrial ecoregions.[15] The CFs were calculated by combining countryside species−area

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relationship (SAR) model[16] with a vulnerability score of the species group.[4]

However, the UNEP-SETAC also recognized that a number of aspects of these CFs need improvement.[14] As such, it recommended that until these improvements are made, the method should only be used for identification of potential impact hotspots but not for product comparison and labeling purposes. The recommended improvements were:

1. First and most importantly, expand land use classes and include different management regimes. In other words, the CFs should be able to differentiate between biodiversity damage caused by different land use classes (e.g., pasture vs cropland) and also between products coming from same land use class but varying in intensity of use (e.g., extensive/intensive pasture).
2. CFs for plants should be calculated and included in the impact assessment.
3. The uncertainty in the CFs, which is currently very high, should be reduced.
4. The model predicted species loss should be compared against the global rate of extinction and/or the observed species loss within the studied systems.
5. Case studies should be conducted to examine the robustness and ability of the model to rank alternative product systems in terms of their biodiversity impacts.

In this study, we present first attempts toward addressing the above five issues and calculate updated CFs for five broad land use types and their three intensity levels.

■ MATERIALS AND METHODS

Countryside SAR Model. We calculate the projected species loss for each species group (g = 1:5; mammals, birds, amphibians, reptiles, and plants) in each of the 804 terrestrial ecoregions (j = 1:804) due to current land use through the countryside SAR model:[16]

$$S_{\mathrm{loss},g,j} = S_{\mathrm{org},g,j} \cdot \left(1 - \left(\frac{A_{\mathrm{new},j} + \sum_{i=1}^{16} h_{g,i,j} \cdot A_{i,j}}{A_{\mathrm{org},j}}\right)^{z_j}\right) \quad (1)$$

Here $S_{\mathrm{org},g,j}$ is the total number of species occurring in each ecoregion's area ($A_{\mathrm{org},j}$) before any human intervention, $A_{\mathrm{new},j}$ is the natural habitat area in the ecoregion currently (in m^2), $A_{i,j}$ is the current area of land use type i (i = 1:16; see Supporting Information, Table S1) in m^2, z_j is the SAR exponent for the ecoregion, and $h_{g,i,j}$ is the affinity of the taxon g to the land use type i in ecoregion j. As in Chaudhary et al.,[4] we obtained the species richness of four vertebrate taxa per ecoregion ($S_{\mathrm{org},g,j}$) from WWF Wildfinder database;[17] species richness of plants from Kier et al.,[18] and the z-values per ecoregion (z_j) from Drakare et al.[19] The procedure to calculate the land use areas per ecoregion and the taxon affinity to human land uses differing in terms of intensity of use is described in detail below.

Land Use Areas Per Ecoregion. We follow a two-step approach to calculate the areas of different land use types and their intensity levels in each ecoregion.

Areas of Broad Land Use Types Per Ecoregion. We first derive the total area ($A_{\mathrm{org},j}$), area of remaining natural habitat ($A_{\mathrm{new},j}$), and the areas of four broad human land use types (secondary vegetation, pasture, cropland, and urban) per ecoregion ($A_{i,j}^{\mathrm{broad}}$) by overlaying the ecoregion boundaries with recently available global "land use" map of Hoskins et al.[20] for the year 2005. We chose the Hoskins et al.[20] map for several reasons. First, unlike LADA[21] and ANTHROME[22] maps used by Chaudhary et al.,[4] it is of approximately 10 times higher resolution (30"). Next, it has been validated using independent global data sets[23,24] and is also compatible with global land use harmonization data set.[25] Because of the above reasons, it is expected to be more accurate than the former. Hoskins et al.[20] has also been the first choice in several recent global biodiversity impact assessment studies.[26−29] Although more recent and high resolution maps of individual land uses such as cropland[30] or forests[31] are available in isolation, to date, Hoskins et al.[20] remains the only global high resolution, harmonized, and validated map providing areas of primary habitat and four broad human land use types as a one consistent data set.

Areas of Broad Land Use Types under Three Intensity Levels. Next, we utilize the land use intensity information from Global Land Systems data set (year 2000)[32] and Arets et al.[33] in order to derive the proportion ($p_{i,j}^{\mathrm{intensity}}$) of broad land use types (secondary habitat, pasture, cropland, and urban land use) under three different intensity levels (minimal, light, and intense use) per ecoregion (Supporting Information, Table S1). Here we assume that land use intensity did not change significantly between the year 2000 and 2005 (year of Hoskins et al.[20] map). The area of a particular broad land use type under a particular intensity level in an ecoregion j ($A_{i,j}$) is then simply calculated as

$$A_{i,j} = A_{i,j}^{\mathrm{broad}} \times p_{i,j}^{\mathrm{intensity}} \quad (2)$$

As an illustration, consider the ecoregion Biak−Numfoor Rain Forests [WWF ecoregion code: AA0103] on the island of New Guinea.[17] According to the Hoskins et al.[20] map, the total area of this ecoregion ($A_{\mathrm{org},j}$) is 2793 km^2, the remaining natural habitat area ($A_{\mathrm{new},j}$) is 2047 km^2, and the area of cropland use ($A_{\mathrm{cropland},j}^{\mathrm{broad}}$) is 168 km^2. Now according to Global Land Systems map,[32] the proportion of cropland ($p_{\mathrm{cropland},j}^{\mathrm{intensity}}$) under minimal, light, and intense use in this ecoregion is 0.14, 0.58, and 0.28, respectively. Therefore, by eq 2 above, the area of cropland under the minimal use is 168 × 0.14 = 23 km^2. Similarly, area of light and intense use cropland is equal to 168 × 0.58 = 98 km^2 and 168 × 0.28 = 47 km^2, respectively.

We consider "minimal use secondary vegetation" to have been abandoned and therefore to be currently regenerating and not used for production (following definition of Newbold et al.[26]). The "light" and "intense" use levels of secondary vegetation in the forest biomes are considered as "managed forests" class, which is further divided into four intensity classes: clear-cut, selective logging, plantations, and reduced impact logging based on their fraction in different world regions as reported by Arets et al.[33] that in turn are based on the Integrated Model to Assess the Global Environment (IMAGE) output.[34] Similar to Newbold et al.,[26] we assumed that the plantation area per ecoregion was composed of equal proportions under minimal, light, and intense use. This is because none of the Global Land Systems[32] classes could inform about the intensity of plantation forestry. We divided the secondary forest vegetation further into four different well-known classes (plantations, clear-cut, selective logging, and reduced impact logging) because these are the major wood production systems across the globe.[35] Separate characterization factors for these systems are therefore needed to differentiate the impacts caused by them.

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Finally, this yielded the areas of 16 land classes, i.e., "regenerating secondary vegetation" plus areas of five broad land use types (managed forests, plantations, pasture, cropland, and urban) under three intensity levels each (minimal, light, and intense use) (see Table S1 in Supporting Information for more details on land use classification).

Taxon Affinity Calculation. As a proxy for taxon affinity ($0 < h_{g,i,j} < 1$) to different human land uses, past studies (including Chaudhary et al.[4]) have relied on published literature documenting the ratio of observed species richness in a particular land use type to that in a natural undisturbed area, obtained from few plot-scale field monitoring within the region.[4,16,36] However, there are large gaps in our knowledge of how damaging a particular land use type is to different taxa in different parts of the world. The uncertainty in h estimates is therefore substantial, and although global field monitoring efforts are underway, these gaps are unlikely to be filled in near future.[4] For example, Chaudhary et al.[4] derived the affinity of five taxa to six land use types in 14 biomes[15] through a meta-analysis of 861 field monitoring data points comparing species richness in human land use and nearby natural (primary) vegetation. For most taxa, land use, and ecoregion combinations, there were no field study available and therefore a biome-average value was taken as a proxy. Furthermore, the data represented only a fraction of the total number of species of a taxa present globally (e.g., the affinity values for mammals, birds, and amphibians were just based on 66, 211, and 11 data points globally).[4] This rendered the affinity values with large uncertainty intervals.

Here, we therefore apply a novel hybrid two-step approach to calculate the affinity of taxa (for which the data is available, i.e., mammals, birds, and amphibians) to different land use types and their intensity levels.

Taxon Affinity to Broad Land Use Types. We first calculate the affinity of taxa to five broad land use types (managed forests, plantations, pasture, cropland and urban) in each ecoregion through the recently proposed approach by Chaudhary and Brooks,[28] i.e., using the species-specific information on habitat utilization documented through the IUCN Red List Habitat Classification Scheme.[37] For each of the 22386 species of these three taxa, the IUCN Habitat Classification Scheme provides information on the human land use types which a particular species utilizes. This information is collected on each species through field observation and literature synthesis by the Red List assessors of the species in question (see section 2.4.1, page 24, of IUCN's documentation standards and consistency checks for species accounts).[38]

For example, the house sparrow (*Passer domesticus*) utilizes all broad human land use types. The frog *Adenomera martinezi* utilizes agriculture and pasture but not in degraded forests, urban areas, and plantations. The primate *Arctocebus calabarensis* utilizes degraded forests only and cannot survive in any other human land use.[37] The species utilization of a particular land use type depends upon its biophysical and life history traits such as diets, body size, temperature range, etc.

We obtained the list of all species of mammals, birds, and amphibians present in each of the 804 ecoregions from WWF Wildfinder database.[17] Next, for each terrestrial ecoregion j, we counted the number of species of each taxon g for a particular land use type i (e.g., cropland) was coded as suitable in the Habitat Classification Schemes of the IUCN Red List.[37] The number of species for which a given habitat was suitable were divided by total number of species of that taxon occurring in the ecoregion, to obtain the fractional species richness ($r_{g,i,j}$). The affinity of taxonomic group g to the broad land use type i is then calculated as the proportion of all species that can survive in it (fractional richness), raised to the power $1/z$ (see Pereira et al):[16]

$$h_{g,i,j}^{\text{broad}} = \left(\frac{S_{\text{org},g,i,j}}{S_{\text{org},g,j}}\right)^{1/z_j} = (r_{g,i,j})^{1/z_j} \tag{3}$$

Taxon Affinity to Land Use Intensity Types. The IUCN Habitat Classification Scheme only provides data on species utilization of five broad human land uses: plantations, agriculture, pasture, urban areas, and degraded forests. To derive taxon affinity to land use intensity classes ($h_{g,i,j}$), we therefore combined the calculated affinity to broad land use types (eq 3) with information from existing meta-analysis studies,[26,35] documenting species richness in different land use intensity levels relative to each broad land use class (fractional relative richness, f_{RR}). The taxon affinity to a land use intensity type ($h_{g,i,j}$) is calculated from the affinity to broad land use type as

$$h_{g,i,j} = h_{g,i,j}^{\text{broad}} \cdot (f_{\text{RR},g,i})^{1/z_j} \tag{4}$$

Here the fractional relative richness ($f_{\text{RR},g,i}$) is the local species richness in a land use intensity type (e.g., intense use pasture) divided by the average local species richness in a broad land use type (e.g., pasture). The mean and 95% confidence interval for the $f_{\text{RR},g,i}$ value for minimal, light, and intense use pasture, plantations, cropland, and urban land uses were obtained from Newbold et al.[26] The data for fractional relative richness of the three managed forest intensity classes (reduced-impact logging, selective logging, and clear-cut) was obtained from the meta-analysis of Chaudhary et al.[35]

For example, the forestry meta-analysis of Chaudhary et al.[35] showed that the average fractional species richness of amphibians in the broad land use class "managed forests" is 0.89 while their fractional species richness in forests managed under clear-cut (intense use) regime is 0.544. Therefore, the fractional relative richness (f_{RR}) value is equal to 0.544/0.89 = 0.6. In other words, the local species richness in "intense use" (clear-cut) forests is just 60% of the species richness value in broad land use "managed forests".

For reptiles and plants, the IUCN Habitat Classification Scheme[37] does not yet provide data for all species, and therefore we derive the affinity values solely using the fractional species richness from Newbold et al.[26] The affinity of taxa to "regenerating secondary vegetation" (under no or minimal use) was also taken directly from Newbold et al.[26]

As an illustration, again consider the ecoregion Biak–Numfoor Rain Forests [WWF ecoregion code: AA0103] that hosts three amphibian species: *Papurana papua*, *Cornufer papuensis*, and *Nyctimystes infrafrenatus*. According to IUCN Habitat Classification Scheme,[37] out of these three species, only one (*Nyctimystes infrafrenatus*) utilizes pastureland. Being an island, the z-value for this ecoregion is 0.44. Therefore, the affinity of amphibians to broad land use "pasture" in this ecoregion according to eq 3 is $h_{\text{amphibian,pasture, AA0103}}^{\text{broad}} = (1/3)^{1/0.44} = 0.08234$. Now the fractional relative richness (f_{RR}) value for "intense use" pasture is 0.8624 as obtained from Newbold et al.[26] Therefore, the affinity of amphibians to "intense use pasture" in this ecoregion is simply: $0.08234 \times (0.8624)^{1/0.44} = 0.0588$ (see eq 4).

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Model Parametrization and Uncertainty Assessment. Finally, the newly calculated land use areas ($A_{i,j}$, eq 2) and taxon affinity estimates ($h_{g,i,j}$, eq 4) were fed into the countryside SAR model (eq 1) to calculate the projected species loss ($S_{\mathrm{loss},g,j}$) per ecoregion. Table 1 presents the data sources and uncertainty distribution of all model parameters. As evident from Table 1, we consider the uncertainty in as many model parameters as by Chaudhary et al.[4] (i.e., in fractional richness and z-values).

For the z-values (z_j) and fractional relative richness ($f_{\mathrm{RR},g,i}$), the mean as well as 95% confidence intervals were available from the respective sources (Drakare et al.,[19] Newbold et al.,[26] Chaudhary et al.[35]) and therefore were used to calculate the mean and 95% confidence intervals for the species loss ($S_{\mathrm{loss},g,j}$) through eq 1. We acknowledge that further uncertainty in taxon affinity to broad land use type ($h^{\mathrm{broad}}_{g,i,j}$, eq 3) is introduced by classification error in application of the IUCN Habitat Classification Scheme;[37,38] no mechanism currently exists to measure this uncertainty, but we are confident that it is small compared to uncertainty in the z-values (z_j) and fractional relative richness ($f_{\mathrm{RR},g,i}$).

No uncertainty information was available for our base land use map (Hoskins et al.[20]) and the maps used to calculate the broad land use types under different intensity levels.[32,33] This was also the case for Chaudhary et al.[4] because the LADA[21] and ANTHROME[22] maps used by them also did not provide uncertainty estimates.

Overall, our approach of first obtaining the area of a broad land use class (e.g., cropland) from a base map[20] and then dividing it into additional land use intensity classes based on their proportion available from another land use maps[32,33] (eq 2) has, to our knowledge, two precedents. For example, Chaudhary et al.[4] used LADA[21] and ANTHROME[22] maps to first calculate the areas of four broad land use types per ecoregion. Chaudhary et al.[4] then divided the total "managed forest" area per ecoregion into "extensive forestry" and "intensive forestry" based on FAO's global forest assessment report data[40] on proportion of forestry under intensive and extensive regime in each country. Similarly, they divided the total "cropland" area per ecoregion into "annual" and "permanent" crop based on the FAOSTAT database.[39] This way they arrived at areas of six broad human land use types per ecoregion (annual crops, permanent crops, pasture, urban, intensive forestry, extensive forestry). Newbold et al.[26] also followed a similar approach by overlaying the land use harmonization map of Hurtt et al.[25] with the Global Land Systems map[32] to calculate areas of broad land use types under three intensity levels per 0.5° grid cell.

Model Validation. Even though countryside the SAR model has been validated many times in the past,[16,28] here we use new land use and taxon affinity data sources[26,35,37] to parametrize the SAR model and therefore it needs to be validated again in order to make sure that the species extinctions predicted by it in each ecoregion compare well against the number of species threatened with extinction.

To validate the parametrized model's extinction predictions, we followed a widely used approach,[2,16,28,36,47,48] comparing the number of endemic extinctions per ecoregion projected by countryside SAR with documented ("observed") number of ecoregionally endemic mammals, birds, and amphibians' species assessed as globally extinct or threatened with extinction on the IUCN Red List,[44] i.e., number of endemic vulnerable, endangered, critically endangered, extinct in the wild, and

Table 1. Uncertainty Distribution of the Model Parameters along with Their Sources Used in This Study[a]

model parameter	uncertainty information	data source (this study)	data source in Chaudhary et al.[b4]
$S_{\mathrm{org},g,j}$ (species richness per ecoregion)	none	WWF Wildfinder database,[17] Kier et al.[18]	WWF Wildfinder database,[17] Kier et al.[18]
$A_{\mathrm{org},j}$ (total ecoregion area)	none	Hoskins et al.[20]	LADA[21] and ANTHROME[22]
$A_{\mathrm{new},j}$ (remaining natural habitat area)	none	Hoskins et al.[20]	LADA[21] and ANTHROME[22]
$A^{\mathrm{broad}}_{i,j}$ (area of four broad land use types−secondary vegetation, pasture, cropland and urban)	none	Hoskins et al.[20]	LADA[21] and ANTHROME[22] maps for areas of four broad land use types
$p^{\mathrm{intensity}}_{i,j}$ (proportion of total broad land use area under a particular intensity level (minimal, light or intense use)	none	Van Asselen and Verburg,[32] Arets et al.[33]	FAOSTAT[39] for proportion of annual and permanent crops per ecoregion and FAO-FRA[40] for proportion of extensive and intensive forestry
$h^{\mathrm{broad}}_{g,i,j}$ (taxon affinity to broad land use type)	none	IUCN Habitat Classification Scheme[37]	Meta-analysis of de Baan et al.,[41] Elshout et al.,[42] Aronson et al.[43] used to calculate fractional richness ($r_{g,i,j}$), eq 3
$f_{RR,g,i}$ (fractional relative richness)	mean and 95% interval considered	Newbold et al.,[26] Chaudhary et al.[35]	
z_j (z-values)	mean and 95% interval considered	Drakare et al.[19]	Drakare et al.[19]
$\mathrm{VS}_{g,j}$ (vulnerability scores)	none	IUCN,[44] Birdlife International,[45] Chaudhary et al.[46]	IUCN,[44] Birdlife International,[45] VS of plants not calculated

[a]The subscripts g, i, j correspond to taxon (g = 5), land use type (i = 15), and ecoregion (j = 804) respectively. The data sources used in Chaudhary et al. for each model parameter are also listed for comparison. [b]In Chaudhary et al.,[4] a median of areas calculated using LADA[21] and ANTHROME[22] maps was used to calculate median species loss per ecoregion. The median and 95% confidence interval for taxon affinity to six land use types per ecoregion was calculated using the meta-analysis of published field monitoring studies.

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Table 2. Average Taxon Affinity ($h_{g,i,j}$; eq 4) of Mammals (M), Birds (B), Amphibians (A), Reptiles (R), and Plants (P) to Different Land Use Intensity Classes (Averaged over 804 Terrestrial Ecoregions)[a]

broad land use	land use intensity	M	B	A	R[b]	P[b]	% area
managed forests	minimal use (RIL)	0.557	0.988	1.000	1.000	0.987	0.4
	light use (selective logging)	0.013	0.036	0.057	0.668	0.607	13.3
	intense use (clear-cut)	0.010	0.013	0.037	0.507	0.686	86.3
plantations	minimal use	0.004	0.018	0.021	0.333	0.333	33.3
	light use	0.003	0.015	0.016	0.259	0.259	33.3
	intense use	0.002	0.010	0.011	0.165	0.165	33.3
pasture	minimal use	0.053	0.020	0.079	0.507	0.507	0.5
	light use	0.044	0.016	0.066	0.388	0.388	91.4
	intense use	0.034	0.012	0.050	0.290	0.290	8.1
cropland	minimal use	0.102	0.115	0.127	0.425	0.425	22.1
	light use	0.076	0.083	0.101	0.299	0.299	36.2
	intense use	0.071	0.077	0.094	0.279	0.279	41.6
urban	minimal use	0.086	0.036	0.104	0.890	0.890	61.7
	light use	0.040	0.015	0.057	0.319	0.319	19.2
	intense use	0.022	0.009	0.031	0.165	0.165	19.2

[a]See Supporting Information, Table S1 for details on land use classification definitions and Tables S3 and S4 for all characterization factors (CFs) per ecoregion, taxa, and land use type. RIL is reduced impact logging. The global % of a particular land use type under minimal light and intense use is also shown. [b]The affinity of reptiles were derived solely from fractional richness data available from the meta-analysis of Newbold et al.[26] For plants the affinity to managed forest classes were derived from meta-analysis of Chaudhary et al.,[35] whereas affinity to all other land use was taken from Newbold et al.[26] Affinity values for mammals, birds, and amphibians were derived by combining IUCN habitat classification scheme[37] with Newbold et al.[26] (see eqs 3 and 4 methods).

extinct species per ecoregion. To derive the projected endemic extinctions per ecoregion, we replaced species richness ($S_{org,\,g,j}$) in eq 1 with number of endemic species per ecoregion ($S_{end,\,g,j}$) (see Chaudhary and Brooks[28] for details on model validation).

To evaluate the model performances against observed extinction risk, we calculated three goodness of fit metrics: root-mean-square error (RMSE), coefficient of determination (R^2), and Nash−Sutcliff efficiency (E_{NS}) between projected and documented values.[28] These metrics tell how close are the model "predicted" values to the "observed" extinction risk values. A lower RMSE and higher R^2 and E_{NS} values indicate better model performance. We also calculated above metrics for the projected extinctions calculated by Chaudhary et al.[4] and compared them with those from this study.

Note that compilation of the IUCN Habitat Classification Scheme[37] is indeed required documentation as part of undertaking the assessment of a species for the IUCN Red List.[44] However, the data that we use for validation (extinction risk categories) here are not informed by the Habitat Classification Scheme,[37] so while the data source is the same (IUCN Web site), the two data sets are independent.

Calculation of Characterization Factors. We here briefly summarize the methodology to calculate CFs giving potential biodiversity damage per m^2 for different land use in different ecoregions (see Chaudhary et al.[4] for full details). The total projected species loss in each ecoregion calculated in eq 1 is allocated to each individual land use type based on their area share and the taxon affinity to them through an allocation factor $a_{i,\,j}$, such that $0 < a_{i,\,j} < 1$ and $\sum_{i=1}^{16} a_{i,j} = 1$:

$$S_{\mathrm{loss},g,i,j} = S_{\mathrm{loss},g,j} \times a_{i,j} \tag{5}$$

$$a_{i,j} = \frac{A_{i,j}(1 - h_{g,i,j})}{\sum_{i=1}^{16} A_{i,j}(1 - h_{g,i,j})} \tag{6}$$

Equation 5 can thus be used to calculate which land use type in a particular ecoregion is responsible for how many species extinctions.[28]

The land occupation CFs giving regional species loss (called *regional CFs* from here on) are then calculated as projected extinctions that can be attributed to an unit area of a land use type i:[2,4]

$$CF_{\mathrm{regional,occ},g,i,j} = \frac{S_{\mathrm{loss},g,j} \times a_{i,j}}{A_{i,j}} = \frac{S_{\mathrm{loss},g,i,j}}{A_{i,j}} \tag{7}$$

Finally, global characterization factors are obtained by multiplying regional CFs with the vulnerability score ($0 \leq \mathrm{VS}_{g,j} \leq 1$) of taxon g in ecoregion j:[4]

$$\mathrm{CF}_{\mathrm{global},g,i,j} = \mathrm{CF}_{\mathrm{regional},g,i,j} \cdot \mathrm{VS}_{g,j} \tag{8}$$

The vulnerability score was calculated by Chaudhary et al.[4] as the summed proportion of the range size for each species occurring in an ecoregion and weighted by their category of extinction risk, from the IUCN Red List.[44] The VS is 1 if all species within an ecoregion are listed as "critically endangered" by the IUCN Red List[44] and have 100% of their range inside it (i.e., strictly endemic).

By combining regional CFs (directly from SAR model) with the vulnerability score, the global CFs intend to capture different aspects of biodiversity damage caused by a product in one single indicator. For example, while the regional CF from the SAR (eq 7) reflect the number of species a product is projected to drive to extinction from that ecoregion only (regional extinctions), the vulnerability score flags the products

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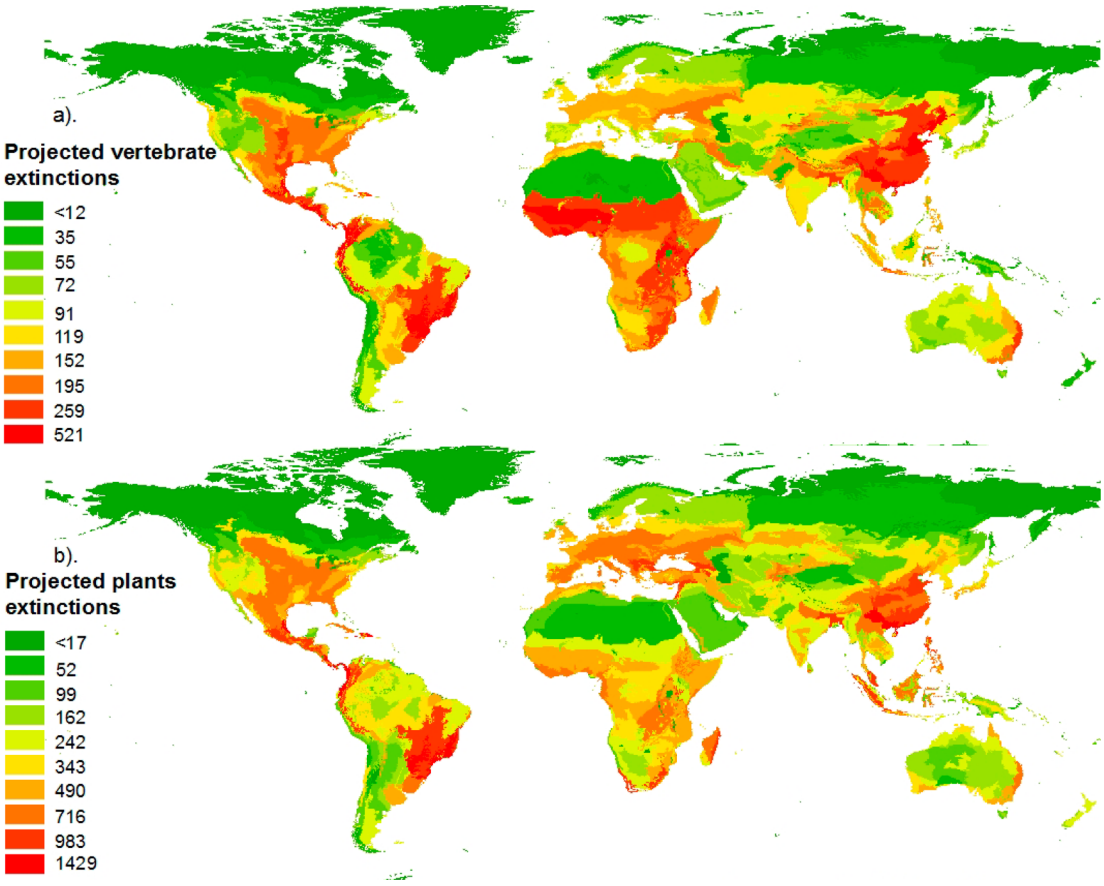

Figure 1. (a) Projected species loss per ecoregion calculated using countryside SAR (eq 1) for four vertebrate taxa combined (mammals + birds + amphibians + reptiles). (b) Projected species loss for plants calculated using eq 1. Figure was drawn in ArcGIS version 10.2.[52]

sourced from areas hosting a high number of threatened and endemic species.

In this study, we imported the vulnerability scores of plants per ecoregion from Chaudhary et al.,[46] enabling us to calculate the global CFs for plants as well (see Table S2 in Supporting Information for all vulnerability scores). The global CFs give a measure of potential global extinctions rather than actual species extinctions, and thus their unit is *potential global species loss per m*^{2}. Note that the CFs are calculated for 15 human land use classes and not for the 16th land class "regenerating secondary forests" that are no longer used for production, and are therefore not relevant for product LCA purposes.

Characterization factors calculations using eq 1−8 are referred to as "average approach CFs" within LCA.[4,49] We did not calculate the characterization factors using the marginal approach[49] because the UNEP-SETAC recommended the use of CFs calculated using the average approach only in order to be consistent with the other indicator methods used in LCA (see page 132 of UNEP report).[14]

Next, we calculate the land transformation impacts by multiplying occupation CFs (eq 8) with half the biodiversity regeneration time (see de Baan et al.,[2] Chaudhary et al.,[4] Koellner et al.,[50] Mila i Canals et al.[51]). Further, we calculate country-specific CFs by weighting the ecoregion CFs with the area of each land use type within each country's different ecoregions.[4,28] Note that one can also calculate the CFs for each land use type aggregated across five taxa in the units potentially disappeared fraction (PDF) by different weighting schemes of their choice.[4,6,46]

Application Example. We demonstrate the applicability of the newly calculated CFs by comparing the potential biodiversity damage caused by 1 m^3 of wood sourced from four different forest management regimes (intense use plantations, clear-cut, selective logging, and reduced impact logging) varying in their management intensity and wood yields in India. We chose India as an example because it is one of the few countries where all four forest management regimes are prevalent for wood production.[33] The country average land occupation CFs for each management regime ($CF_{global,\,g,i,c}$) were divided by corresponding country-average wood yields ($Y_{i,c}$ in m^3/m^2) for that regime (taken from Arets et al.[33]) to calculate the potential biodiversity damage ($BD_{global,\,g}$ in potential species loss) for each taxon g:

$$BD_{global,g} = \frac{CF_{global,g,i,c}}{Y_{i,c}} \tag{9}$$

For comparison, we also calculate the impacts using the country-average CFs derived by Chaudhary et al.,[4] which are only able to differentiate between intensive and extensive forestry regimes.

Note that the goal of this simple example is just to show the comparison of land occupation biodiversity impacts due to forestry regimes differing in level of intensity, calculated using the new CFs of this study and the old CFs of Chaudhary et al.[4]

RESULTS

Countryside SAR Model Parametrization. Table 2 shows the global average affinity of each taxon to 15 different

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land use intensity classes calculated in this study (eq 2). We found that in general for a particular land use type, the affinity value derived using species-specific habitat classification scheme[37] were lower than values derived by Chaudhary et al.[4] using the global meta-analysis of field monitoring studies comparing local species richness in natural habitat and human land use types.[41−43]

For example, the average affinity of mammals to the broad land use class "pasture" is an order of magnitude smaller (∼0.04) when calculated using the IUCN Habitat Classification Scheme[37] compared to the corresponding value of 0.40 documented through meta-analysis[26] (Table 2). Similarly, whereas the meta-analysis Newbold et al.[26] found that fractional species richness ($r_{i,g,j}$, eq 3) averaged over all taxa in "urban" land is ∼60% of those found in natural habitat of the same region, the IUCN Habitat Classification Scheme data shows that on average just 20% of all species in an ecoregion utilize urban land use.

Compared with managed forests and urban land use classes, for the other three broad land use classes, we found that the affinity of taxa to minimal, light, and intense use differed relatively less (<factor of 2). The response of different taxa to a particular land use type also showed a variation less than a factor of 2.

Table 2 also shows the % of a particular land use type under minimal, light, and intense use currently globally. Thus, ∼85% of the total managed forest and 42% of total cropland area globally is under intensive use. For pasture, light use is the most common intensity level (>90%). In the biodiversity context, <1% of global forestry and pasture land are under "minimal use", which is least damaging to local species richness.

Figure 1 shows the number of species projected to be driven to extinction due to current human land use in each ecoregion as calculated by our parametrized countryside SAR model (eq 1). For both the vertebrates and plants, many extinctions are projected in the ecoregions that host a high number of species and where most natural habitat has been converted to human land use.

Model Validation. We found that our SAR projected endemic species extinctions for each taxon per ecoregion compare well with the number of ecoregionally endemic mammals, birds, and amphibians documented as threatened with extinction by the IUCN Red List.[44] Table 3 shows that the coefficient of determination (R^2), Nash−Sutcliff coefficient of efficiency (E_{NS}), and Spearman rank correlation (ρ) values calculated by comparing projected and documented extinctions were much higher (∼0.6−0.9) for the model parametrized in this study using the newly calculated land use areas and taxon affinity estimates as compared to the model parametrized by Chaudhary et al.,[4] thereby showing an improved model performance and more accurate characterization factor values.

In absolute terms, the extinctions projected per ecoregion by Chaudhary et al.[4] were generally lower than our estimates. This was because the taxon affinity estimates derived by Chaudhary et al.[4] through literature review[41−43] were higher than those calculated using species-specific habitat classification scheme[37] in this study.

Updated Characterization Factors. Table S3 in Supporting Information presents the newly calculated land occupation characterization factors (CFs) for each taxa, land use type, and ecoregion using the "average approach" (potential species loss per m^2, eq 8). Supporting Information, Table S4, shows the corresponding land transformation CFs, while Tables S5−S6 show the country-aggregated CFs.

Table 3. Model Validation Results: Goodness of Fit Metrics and Spearman Rank Correlation Coefficient (ρ) Calculated by Comparing Countryside SAR Model Predicted Endemic Extinctions with Number of Species Documented As Threatened with Extinction on the IUCN Red List in 804 Terrestrial Ecoregions[a]

taxon	metric	this study	Chaudhary et al.[4]
mammals	RMSE	0.62	6.61
	R^2	0.79	0.59
	E_{NS}	0.66	0.26
	ρ	0.76	0.35
birds	RMSE	0.70	5.44
	R^2	0.64	0.55
	E_{NS}	0.61	0.50
	ρ	0.86	0.21
amphibians	RMSE	1.72	11.32
	R^2	0.89	0.79
	E_{NS}	0.80	0.67
	ρ	0.75	0.54

[a]A lower root-mean-square error (RMSE) and a higher coefficient of determination (R^2), ρ, and Nash−Sutcliff coefficient of efficiency (E_{NS}) values indicate better model performance.[28]

We found that for a particular taxon and land use type, the CFs varied by ∼4 orders of magnitude across 804 ecoregions with higher values in species-rich tropics and lowest values in boreal ecoregions (Table S3 in Supporting Information). For a given taxon, the CFs varied within 1 order of magnitude across the 15 land use types for most ecoregions. For a given ecoregion and taxon, plantations, agriculture, and urban land use classes had in general the highest CFs, reflecting the relatively low affinity ($h_{g,i,j}$, eq 3) of taxa to them as compared to forestry or pasture land uses. Within the forestry land uses, the CFs were generally in the order: clear-cut > selective logging > reduced impact logging. The differences between the CFs for agriculture and pasture intensity classes were less pronounced. These trends reflect the relative order of taxon affinities to different land use intensity types (Table 2).

We found that for a given land use type, the new CFs are in general higher than the CFs calculated by Chaudhary et al.[4] mainly because their taxon affinity estimates were higher than those calculated in this study. Importantly, the 95% confidence intervals of our new CF values are significantly narrower than the CFs of Chaudhary et al.,[4] which ranged from negative to positive. Finally, unlike the CFs of Chaudhary et al.,[4] the new CFs enable comparison of products coming from three different management intensity levels (minimal, light, and intense).

For the newly calculated CFs to be operationalized into the LCA software, we also provide a look up table that matches the land use intensity categories used in this study with the land use intensity classes recommended by the UNEP-SETAC for use in LCA (Koellner et al.[52]) and also used in LCA inventory database such as ecoinvent v3 (Supporting Information, Table S7).

Application Example. Figure 2 shows that the new characterization factors (CFs) are better able to discern the impact of different forest management types as compared to the

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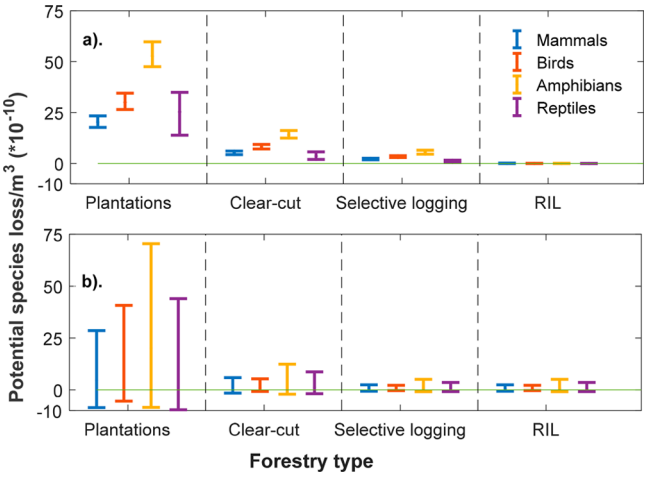

Figure 2. Biodiversity damage caused by 1 m^3 of wood sourced from four different forest management regimes in India calculated by using (a). New characterization factors derived in this study (b). Characterization factors derived by Chaudhary et al.[4] Except timber plantations, the other three management regimes fall into "extensive forestry" category of Chaudhary et al.[4] See Supporting Information, Table S8 for detailed calculations (including plants) and data sources.

old CFs provided by Chaudhary et al.[4] It can be seen that using the Chaudhary et al.[4] CFs for each taxa, the calculated 95% confidence interval of the potential species loss values under all four regimes overlap with each other as well as cross the CF = 0 horizontal line. This means that one cannot conclude if the calculated damage is significantly different from zero (statistically) or if the wood sourced from any one regime is less damaging than the other.

In contrast, the potential species loss for any given taxa, calculated using the new CFs, clearly show that wood sourced from plantation forest causes the most damage followed by that sourced from clear-cut, selective logging, and reduced impact logging (RIL) regimes because the 95% confidence interval under a particular regime neither overlaps with the other three nor intersects the CF = 0 line (except RIL).

For example, using the new CFs, the calculated impact due to land occupation associated with producing a 1 m^3 of wood in a selectively logged forest and a forest managed using RIL is equal to 2.04×10^{-10} and 7.09×10^{-12} potential species loss, respectively (Figure 2; Supporting Information, Table S8). In other words, the land occupation impact is 30 times lower if the wood is sourced from RIL forest compared to selectively logged forest. However, the calculated impact using the old CFs of Chaudhary et al. is equal to 5.19×10^{-11} for both selectively logged and RIL forest. This is because both of these forestry regime fall under the "extensive forestry" definition of Chaudhary et al.[4]

Therefore, the simple example above shows that new CFs are able to differentiate the impact caused by different land use intensity classes, something that was not possible with old CFs.[4] Note that here we only compared the land occupation impacts of the four forestry regimes but the ranking of total impact might change if other factors such as land use change, fuel use etc. associated with each regime are taken into account.

■ DISCUSSION

The study presents first attempts toward addressing the gaps identified by UNEP-SETAC life cycle initiative regarding calculation of land use biodiversity impacts within life cycle assessment (LCA).[14] First and most importantly, by using the global land use and land use intensity maps,[20,32,33] we derived the new characterization factors (CFs) that enables one to discern the damage caused by three levels of intensity classes within a particular broad land use type, something not possible through the CFs of Chaudhary et al.[4]

Note that the UNEP-SETAC recommended to both expand the number of land use classes as well as include different management regimes in the biodiversity model (see page 139 of UNEP report).[14] While we do not improve upon the number of land use classes, we do include three management regimes (minimal, light, and intense use) for each broad land use class that were absent from Chaudhary et al.[4] Second, we derive CFs for plants.

Third, for calculating taxon affinity to different land use intensity types, we applied a novel hybrid approach of leveraging Habitat Classification Scheme from the IUCN Red List[37] and combining it with empirical data on species richness comparisons from literature.[26,35] This enabled us to bring down the uncertainty range of the CFs within a factor of 2 (Figure 2; Supporting Information, Tables S3−S6). Previous CFs from Chaudhary et al.[4] had huge uncertainty range varying over several orders of magnitude (from positive to negative) and thereby limiting their practical utility.

Fourth, model validation and goodness of fit metrics (Table 3) showed that extinction projections by our parametrized model compare very well with the number of species documented as threatened with extinction by the IUCN Red List[44] and much better than the model parametrized by Chaudhary et al.[4]

Finally, we demonstrate the application of new CFs through a case study on comparing the land occupation biodiversity impacts of 1 m^3 of wood sourced from four different forest management regimes (Figure 2).

The innovations presented here do not resolve all issues identified by the UNEP[14] but rather are steps forward in advancing the field. The main novelty of this study is the hybrid approach of combining the Hoskins et al.[20] map with a land use intensity data sets[32,33] to calculate the areas of five broad land use types under three intensity levels (eq 2, Table 2) and combining the IUCN Habitat Classification Scheme[37] with data from meta-analyses[26,35] on fractional richness to calculate affinity of taxon to the land use types under different intensity levels (eq 3, 4). Previous studies (Chaudhary and Brooks[28]) used the Hoskins et al.[20] map to calculate the areas of four broad human land use types per ecoregion and the IUCN Habitat Classification Scheme[37] to calculate the affinity of mammals, birds, and amphibians to these four land use types only.

Our intermediate results on the calculation of taxa affinity to different land use intensity types (Table 2) and projected extinctions per ecoregion (Figure 1a, b) provide important insights for biodiversity conservation and policy making. For example, we found that taxon affinity values differ substantially across the three intensity levels for managed forest land use (Table 2). This implies that shifting from minimal use forestry regimes (reduced impact logging) to light/intense use forestry (selective logging/clear-cut) would cause substantial additional losses. This has implications for regions planning to intensify their production systems or, conversely, to shift them toward biodiversity friendly management practices.

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Despite significant improvements, substantial uncertainties in accurately predicting the biodiversity damage caused by a particular land use in different parts of the world remain due to several data gaps, model, and parameter uncertainties and the value choices involved in the characterization approaches. Where available, such as in case of z-values[19] and fractional relative richness[26,35] (f_{RR}, eq 4), we propagated the uncertainty in model parameters (Table 1) into the characterization factors. However, in many cases, it was not possible because the underlying data sets do not report the uncertainty intervals around their estimates such as in the case of the land use and land use intensity maps we used to derive areas of different land use types under minimal, light, and intense use.[20,32,33]

We acknowledge that our approach of combining two different land use data sets might add uncertainty to the results that could not be quantified by us. We could only consider five broad land use classes and their three intensity levels but future studies should attempt to include further land use classes (e.g., annual, permanent crops, organic farms etc.) Although rough procedure do exist, such as adopted by Chaudhary et al.,[4] to translate country level proportion[39] of annual and permanent crops to ecoregion level proportion, these are bound to add further unknown amount of uncertainty into the results and therefore not applied here. New global high-resolution, harmonized, and validated land use maps with more land use and intensity classes are needed to reduce the uncertainty in characterization factors.

Next, the Habitat Classification Scheme of the IUCN[37] is not yet available for all species of plants and reptiles and is only available for six broad human land uses for mammals, birds, and amphibians. We therefore still had to rely on fractional relative richness factors ($f_{RR,g,i}$, eq 4) from published meta-analyses[26,35] of limited field studies (eq 4). A big limitation of the fractional relative richness factors that we used is that they are taxa and region generic. It is possible that one taxa is more sensitive to an increase in intensity of land use than others or that an increase in intensity affects the species differentially in different ecoregions. This might be the reason that the calculated CFs do not differ much across the three intensity classes for cropland and pasture land use (Supporting Information, Table S3). For example, according to Newbold et al.,[26] the average fractional species richness ($r_{g,i,j}$, eq 3) in "minimal use" and "intense use" cropland is 0.73 and 0.64, respectively, i.e., a difference of just 9%. Consequently, the calculated global CFs for cropland vary <10% across the three intensity classes (Supporting Information, Table S3).

Another reason for a small difference in CF magnitude across the three intensity levels can be that most species are already wiped out once the natural vegetation of a region is cleared to make way for pasture/cropland whether it is then used with minimal, light, or intensive management. The features of natural vegetation (e.g., high canopy cover in tropical ecoregions providing shade, reproduction sites, etc.) are not found in pasture/cropland, and therefore this human dominated habitat is rendered unsuitable for many species. Many other species that can ideally survive in pasture/cropland are not allowed to utilize these habitats because of a conflict for food with the livestock or they pose a threat to livestock/humans.

We simply do not have enough data on a global scale telling us which individual species of each taxa can survive in which intensity class in different world regions. As species-specific habitat classifications for different intensity gradients come along in the future, the approach proposed here can be applied to recalculate the more accurate CFs. For now, at least while comparing the same product from these three intensity gradients, we get slightly different impacts, which is an improvement over previous intensity generic CFs of Chaudhary et al.[4]

We acknowledge that the indicator of biodiversity loss presented here (potential species loss) is just one of the many indicators of global biodiversity loss and the ranking of characterization factors among different ecoregions might change if different model/indicator is selected.

In addition, the several underlying assumptions and limitations of SAR approach in projecting extinctions must be considered while interpreting the results (see ref 28 for details). As more data becomes available, future studies should increase the taxonomic coverage of the analysis (e.g., including invertebrates) and also quantify land use impacts on other indicators of biodiversity (e.g., functional[53] or genetic diversity[29]). Many studies in the recent past have called for improvements in quantitative biodiversity footprinting methods[9−11,54,55] such as ability to discern the impacts of biodiversity friendly production systems from those operating with nonsustainable, profit-driven, intense, and damaging practices.

In our simple case-study application (Figure 2), we did not take into account the total life cycle inventory, potential land transformation impacts, a complex production chain, regions where different processes take place, etc. More complete case studies considering above factors are needed in the future to test the robustness of CFs. The goal of application example was simply to show that by applying separate CFs for different land use intensity types (e.g., selective logging vs reduced impact logging), we get different impacts. In contrast, by using the old CFs of Chaudhary et al.,[4] one would get the same impact for both minimal use (reduced impact logging) and light use (selective logging) forestry regimes in a region.

Apart from product comparison and life cycle assessment applications, the availability of CFs for different land use intensity levels is also valuable for studies quantifying national or regional biodiversity footprints by linking CFs obtained from SAR model with global production, consumption, and trade databases.[8,28,56−59] Working with broad land uses such as forestry, pasture, agriculture, and urban, means that biodiversity footprint of two contrasting countries comes out to be same (one that imports food or wood products produced at a farm managed using minimal use cropland or forestry practices while the other from intensively managed farms).

Application of the new CFs presented here gets countries or businesses one step closer to being capable of quantitatively demonstrating that their biodiversity friendly production regimes have lower impacts than the players operating with relatively intensive production systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.7b05570.

Land use classification; overview of ecoregions, habitat tpes, ecoregion area, species richness, and vulnerability scores of each taxa per ecoregion; global land occupation characterization factors per ecoregion; global land transformation characterization factors per ecoregion;

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country aggregated global land occupation characterization factors; country aggregated global land transpormation characterization factors; matching the land use intensity catergories used in this study with the land use intensity classes recommended by the UNEP-SETAC for use in LCA as well as those used in LCA inventory database; case-study calculations (XLSX)

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Author Contributions

A.C. conceived the idea, designed study, carried out all calculations, and wrote the manuscript. T.M.B. contributed critical feedback and manuscript revision.

Notes

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